



Acoustic and perceptual spatial delimitation in public places - a multi-criteria method

Delimitação espacial acústica e perceptiva em espaços públicos - um método multicritério

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ABSTRACT

The urban space is modulated in its spatial delimitation by qualitative and quantitative criteria of the built environment and the various atmospheres in which it is situated. This scope and extent, in turn, give rise to a delimitation and a notion of environmental marking boundaries, which encompasses the full diversity of social practices and behaviors that are more or less profoundly modulated by the perceptual aspects of the environment. This is the notion of public space in its environmental dimensions. Urban noise can thus be a marking element, a rupture, a link, or an articulation between spaces that are morphologically or visually incompatible. We can even speak of sonic delimitation of urban space beyond its morphological limits, and this delimitation effectively connects the social practice of urban space with its dimensional aspect. A study conducted on the presence of a fountain as a noise source in the center of a public square clearly supports this marking of space and introduces the concept of acoustic space as opposed to morphological space, which must now inevitably be reconciled in order to better design or manage urban entities. Noise helps to highlight the spatial dimension of public space. It even qualifies the space as closed or open and reinforces the visual perception of the place. The elements that compose the atmosphere or



environment must contribute to modifying or requalifying an urban space in its marking and delimitation

Keywords: Marking and Delimitation. Sound Environment. Urban Space. Spatiality. Multi-Criteria. Qualitative Aspects. Social Practices. Acoustic Space.

RESUMO

O espaço urbano é modulado em sua delimitação espacial por critérios qualitativos e quantitativos do ambiente construído e pelas diversas atmosferas em que está inserido. Esse alcance e extensão, por sua vez, dão origem a uma delimitação e a uma noção de fronteiras de marcação ambiental, que abrange a diversidade das práticas sociais e dos comportamentos, mais ou menos profundamente modulados pelos aspectos perceptivos do ambiente. Essa é a noção de espaço público em suas dimensões ambientais. O ruído urbano pode, assim, ser um elemento de marcação, ruptura, ligação ou articulação entre espaços morfologicamente ou visualmente incompatíveis. Podemos até falar de delimitação sonora do espaço urbano além de seus limites morfológicos, e essa delimitação conecta efetivamente a prática social do espaço urbano ao seu aspecto dimensional. Um estudo realizado sobre a presença de uma fonte como fonte de ruído no centro de uma praça pública apoia claramente essa marcação do espaço e introduz o conceito de espaço acústico em oposição ao espaço morfológico, que agora deve ser inevitavelmente reconciliado para melhor projetar ou gerir entidades urbanas. O ruído ajuda a destacar a dimensão espacial do espaço público. Ele qualifica o espaço como fechado ou aberto e reforça a percepção visual do local. Os elementos que compõem a atmosfera ou o ambiente devem contribuir para modificar ou requalificar um espaço urbano em sua marcação e delimitação.

Palavras-chave: Marca e Delimitação. Ambiente Sonoro. Espaço Urbano. Espacialidade. Multicritério. Aspectos Qualitativos. Práticas Sociais. Espaço Acústico.

RESUMEN

El espacio urbano está modulado en su delimitación espacial por criterios cualitativos y cuantitativos del entorno construido y por las diversas atmósferas en las que se sitúa. Este alcance y extensión, a su vez, dan lugar a una delimitación y a una noción de fronteras de marcación ambiental que abarcan la diversidad de prácticas sociales y comportamientos, más o menos profundamente modulados por los aspectos perceptivos del entorno. Esta es la noción de espacio público en sus dimensiones ambientales. El ruido urbano puede, por tanto, ser un elemento de marcación, una ruptura, un vínculo o una articulación entre espacios que son morfológica o visualmente incompatibles. Incluso se puede hablar de una delimitación sonora del espacio urbano más allá de sus límites morfológicos, y esta delimitación conecta efectivamente la práctica social del espacio urbano con su aspecto dimensional. Un estudio realizado sobre la presencia de una fuente como fuente de ruido en el centro de una plaza pública respalda claramente esta marcación del espacio e introduce el concepto de espacio acústico en oposición al espacio morfológico, que ahora debe ser inevitablemente reconciliado para diseñar o gestionar mejor las entidades urbanas. El ruido ayuda a destacar la dimensión espacial del espacio público. Incluso califica el espacio como cerrado

o abierto y refuerza la percepción visual del lugar. Los elementos que componen la atmósfera o el entorno deben contribuir a modificar o revalorizar un espacio urbano en su marcación y delimitación.

Palabras clave: Marcación y Delimitación. Entorno Sonoro. Espacio Urbano. Espacialidad. Multicriterio. Aspectos Cualitativos. Prácticas Sociales. Espacio Acústico.

1 INTRODUCTION

1.1 BACKGROUND

From the perspective of human hearing, sounds are understood as physical phenomena, perceived through variations in air pressure that are detected by the ear. As a result, sound is often treated as a purely physical phenomenon, with sound level measurement being the most commonly used method for assessing sound quality or noise annoyance (HALL *et al.*, 2013). While sound is generated by pressure waves, noise also involves a subjective judgment by the listener (LONG *et al.*, 2005).

Figure 1. Place Notre-Dame in Grenoble: the sound of the fountain takes on its full significance in shaping the experience of the square's space.



(The fountain in Place Notre-Dame in Grenoble-France. Photo by the authors)

“What is the relationship between man and the sounds of his environment [...]? [...] Only a total appreciation of the acoustic environment can give us the



resources for improving the orchestration of the world soundscape” (SCHAFER, 1977). However, over the numerous studies in the field of urban acoustic environments, it has become evident that the human dimension is of great importance in understanding the issue of urban noise. Sound is one of the five major ways that we use to communicate with our surroundings (TAYLOR, 2000).

Thus, a key distinction between the field of sound environment and traditional urban noise studies lies in the central role of human perception (*BROWN et al., 2011*). Social science methods, such as surveys and interviews, complement physical sound measurements to capture the subjective perception of sounds (*PÉREZ-MARTÍNEZ et al., 2018*). These approaches increase awareness of sound environment concepts and enable evaluations from both objective and subjective perspectives (*PORTEOUS et al., 1985*).

In the other hand, a public space is a space often situated within an urban fabric. Whether this space is tangible and physically materialized, or abstract and defined by perceptual experience and shaped by socio-cultural relations, it remains a whole that is difficult to decompose. Its physical and morphological boundaries overlap with its perceptual scope, with social, cultural, and symbolic practices, all of which combine to give it shape and existence. Even visual information of the location affects noise evaluation (*VIOLLON, 2003*). It was concluded that the influence of visual and verbal information on the auditory evaluation of environmental sounds is considerable (*ABE et al., 2006*). The perceived quality of a sound environment is strongly influenced by individual subjective factors, such as personal preferences, past experiences, and social and cultural contexts.

Often, there is a tendency to define a public space by its configuration, accessibility, or spatial and morphological components, thus limiting it to a purely visual representation. However, it is essential to also conceive of public space as an urban entity defined by lived experience and the social activities and practices that occur within it.

We also observe the phenomenon of sonic marking of space, which is very present in the daily experience of public spaces. Generally, noise stems from human activities, particularly urbanization and the expansion of transportation and industry (SINGH, 2004). Various activities and events in urban or public spaces give these spaces a specific sound identity. The tram on a boulevard, the fountain



in a square... are very common in the reality of this type of public space in our cities. Just like the visual, socio-cultural, or other aspects, sound or noise in public space strongly marks its marking boundaries and spatiality by creating a sound environment characteristic of the dimensions and shape of that space. It is now imperative to adopt a multi-criteria approach to urban phenomena in its complexity and diversity.

To evaluate urban soundscape quality, various methods focus on the subjective responses of park users. A study was conducted for a comprehensive assessment of soundscapes in four typical urban open spaces in Sheffield, UK, using semantic differential analysis (KANG *et al.*, 2010). This technique was designed to analyze the emotional connotations of words and has since been adapted for a wide range of concepts (OSGOOD *et al.*, 1957).

In this *methodological context*, it is often accepted that social interactions, which either reinforce or weaken its spatial and morphological or urbanistic aspects, take precedence in shaping the real form of public space. A study was conducted within a broader context specifically focusing on sound environments in urban areas. It was through this study that phenomena of *sonic marking and sonic delimitation of urban spaces* were highlighted. It was also noted that a *concept of delimitation* was established through the social practice and experience of certain public spaces under study.

This paper aims to present certain aspects of that study. A *multi-criteria approach* was used with the objective of better understanding this notion of sonic delimitation of urban spaces. The presence of a prominent sound source in most of the studied sites provided highly favorable conditions for demonstrating such concepts. This paper will also present key case *studies on sonic delimitation* of space through acoustic and perceptual phenomena in urban settings. The study focused on urban sites in the cities of Grenoble (France), Setif, and Constantine (Algeria).



1.2 RESEARCH CONTEXT

1.2.1 Sound component in urban space

In our daily lives, noise is often regarded as a mere auditory phenomenon. However, it is not just the back-and-forth of sound between hearing and the environment; it represents the link that creates a relationship between listening and the environment. This relationship, which often arises through various acoustic and perceptual phenomena, modulates and conditions, particularly through listening experiences, our habitual interaction with the environment. As a powerful carrier of information, sound either reinforces connections or emphasizes breaks. It is an accurate indicator of a specific spatial delimitation and a form of marking boundaries in the social and daily use of space.

It is quite interesting to understand how noise, therefore, conditions the spatiality of public places and defines perceptual sub-spaces, which in turn shape the lived experience of the area. Through this phenomenon of sonic delimitation, urban noise significantly influences interpersonal relationships or those between individuals and their surroundings. It is a means by which the spatial, temporal, and socio-cultural aspects of a place interact, often clearly defined by the sonic environment.

1.2.2 The fountain in public spaces

A portion of this work focused on studying the phenomenon of sonic marking in urban spaces. The concept of spatial delimitation through noise appeared more evident during the study of a sound source that stands out in the urban soundscape.

The fountain represents the most pronounced and elaborate form of this presence. It is also noteworthy how the fountain adds flexibility and articulation to a public square in terms of urban morphology, physical and functional organization, and especially in shaping the atmospheres within the city.

Another aspect of water and fountains in public squares, which seems most relevant to this study, is their sound. It is one of the key qualities of a fountain that



animates and brings life to a square. Through the rhythm and sound composition, the sonority of the fountain can sometimes be more significant than its visual display. Just as with its other aspects—visual, aesthetic, or symbolic—the noise and sound of a fountain strongly mark the space and environment in which it is located. Sound, in this case, can be an element of marking, disconnection, connection, or articulation between spaces that are otherwise difficult to reconcile. A concept of marking boundaries and sonic delimitation of space takes shape and comes into existence around a sound source in an urban space.

1.3 RESEARCH OBJECTIVES

This paper will mainly focus on the presentation of the methodological aspect along with some results and conclusions.

The main objectives are, first, to study the sound marking and acoustic delimitation in urban areas from different perspectives. The study is conducted using multiple analysis criteria, allowing for a deeper understanding of this phenomenon. Another aim is to closely examine the social and daily use of urban space through sound marking and zoning in public spaces. Finally, one of the goals of this study is to develop a multidisciplinary approach to address the phenomenon of sound delimitation, with the eventual aim of controlling urban noise.

2 METHODOLOGY

In all research hypotheses surrounding the concept of spatial delimitation in the perceptual and social practice of public spaces, our approach followed several lines of investigation. Each phase aimed to highlight the parameters conditioning different aspects of the sound phenomenon and to better grasp its manifestations in the environment. Indeed, sound may have effects beyond those classified as acoustic or perceptual. The fountain was considered a central noise source, around which various phenomena related to sound environments in urban areas were shaped and manifested. The investigative stages were as follows:

1. **Acoustic study, metrological measurements**, and identifying forms of manifestation of the sonic marking and delimitation phenomenon.



2. ***Psycho-sociological survey*** to restore the socio-cultural and symbolic dimension of the concept of delimitation and sonic marking. This involved understanding the space-noise-living experience interrelations around the sound source through acoustic and perceptual phenomena.
3. ***Sound recording and sequence analysis*** taken from various sound environments. This further reinforced the effort to identify the phenomenon of sonic marking and the concept of spatial and territorial delimitation already explored through metrology and psycho-sociological surveys.

3 RESULTS AND DISCUSSIONS

3.1 ACOUSTIC STUDY, METROLOGICAL MEASUREMENTS:

3.1.1 Results

A phase of measurements was carried out to account for various aspects of the sound environment in urban sites. Metrology is a necessary, though insufficient, tool for studying the practice of urban space through acoustic and perceptual components. Other dimensions of noise appear to be important in analyzing issues related to the concept or phenomenon of sonic marking in public spaces. Metrology can provide a clear understanding of the phenomenon in question.

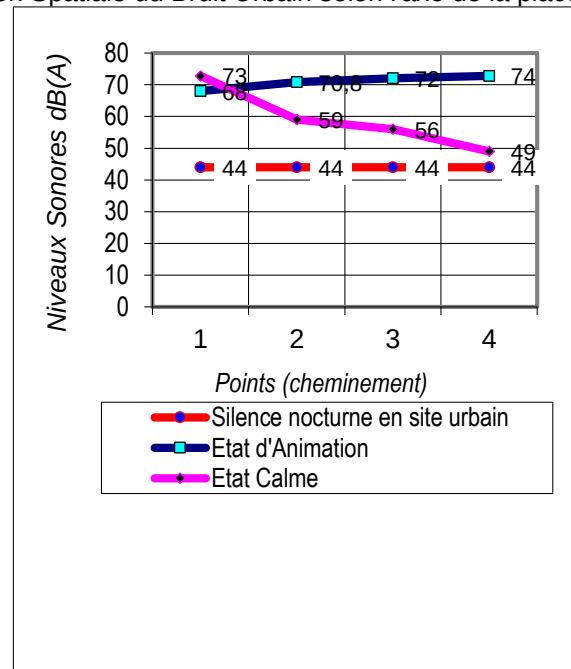
For acoustic measurements, numerous points were designated at each site to conduct this phase of fieldwork. The number and location of these points were primarily dictated by the spatial configuration of the area and the nature of the acoustic aspect or sound phenomenon to be highlighted. A preliminary study of each site helped determine locations aimed at providing a global assessment of the sound environment at different times of day. The presence of cascading or jet fountains in most of the studied public squares required additional measurement points, enabling the spectral analysis of the fountain noise.

3.1.2 Discussion: case study: Place Victor Hugo in Grenoble

In terms of sound, Place Victor HUGO shows a certain uniformity in noise levels across the different measurement points. The square is completely surrounded by roads and urban traffic. The sound of the fountain is softened by the large water surface of its basin and likely by the greenery that surrounds it. However, the richness of the fountain's sound spectrum, particularly in higher frequencies (which is typical of fountains with water jets), allows its sound to spread, albeit with difficulty, across a large portion of the square.

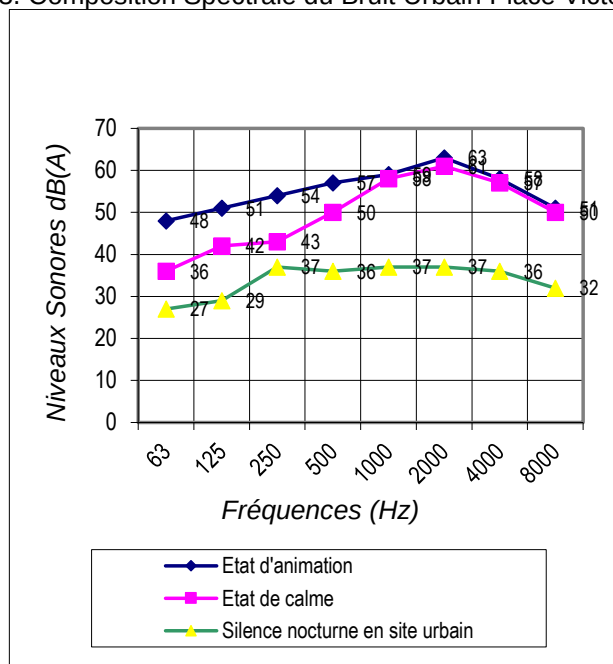
Yet, it remains challenging for the fountain's noise to drown out the steady urban drone, which is rich in low frequencies. As shown in the diagrams, the noise levels of the fountain (*point 1*) and urban traffic (*point 4*) are nearly the same, with the main difference lying in the sound quality at each point (a noticeable difference in frequency texture and timbre is perceived as one moves from the fountain toward the tramway track).

Figure 2. Modulation Spatiale du Bruit Urbain selon l'axe de la place Place Victor HUGO



Source: Authors.

Figure 3. Composition Spectrale du Bruit Urbain Place Victor HUGO



Source: Authors.

If we trace the acoustic path from the fountain to the boulevard (from *point 1* to *point 4*), we observe that apart from *point 1*, which is closest to the fountain, the noise takes on a horizontal character (*during busy periods*), reflecting the constant and enveloping nature of the urban drone. This boundary is quite visible in how people use this space. The farther one moves from the fountain, the more evident this spatial boundary becomes, manifesting as a change in sound quality. Beyond *point 2*, the noise in the square becomes steady and largely devoid of informational value. The layout and especially the use of benches in the square are highly influenced by the sound environment. The acoustic marking is clearly established through the lived experience of the urban space, shaped by perceptual phenomena.

3.2 PSYCHOSOCIOLOGICAL SURVEY

3.2.1 Results

In order to better understand the various components of the sound environment in public squares and spaces, psycho-sociological interviews and surveys were conducted. This aspect of the work complements the multi-criteria



approach used to analyze and accurately assess the scope and impact of the various parameters shaping the soundscape. The goal of this approach is to reflect the lived experience of the square through its sound ambiance and to better grasp the hypothesis regarding the concept of marking boundaries and the acoustic delimitation of urban spaces.

From initial observations made on urban spaces with the presence of certain noise sources such as fountains, trams, or often traffic, it was noted that sound has a significant influence on the daily life and routine of these sites. The interactions between different types of noise and the wide range of sound sources in the urban environment created situations with multiple acoustic and perceptual phenomena. In the expressions of the interviewees, there appeared to be an indication of certain sound-based demarcations and delimitation within these spaces.

A series of interviews based on progressively more specific questions, from the general to the precise, and from factual to more interpretive, was conducted to capture the sometimes complex reality of sound. This survey sought to highlight the daily social experience and use of public spaces, focusing on the phenomenon of delimitation and sound demarcation in urban areas.

The interviews followed structured questions and prompts, progressing in levels of specificity to better understand the subject matter and capture the reality of sound marking and delimitation in the use of public space through urban noise. The aim was to explore the relationship between the spatial morphology of the location and its sound ambiance. Questions were formulated to encourage interviewees to express, in their own words, how they perceive and use the space, particularly in relation to its sound dimension. The focus was on identifying sound marking in public spaces through various acoustic and perceptual phenomena, such as sound masking, reverberation, and disruption. Additionally, many of the interviews aimed to find correlations between what was described by the interviewees and what was measured through metrology.

3.2.2 Discussion: case study: Place Sidi Djellis in the Medina OF Constantine

This square is highly appreciated by its regular visitors. Apart from causing slight discomfort for claustrophobes or those who prefer open spaces, it provides

a sense of relief with its calm and soothing sound, which contrasts with its location in the noisy urban center. The fountain, slightly off to the side, effectively covers this nearly enclosed space (*it's rare to find such squares. This one is well placed, right in the center, which I believe is the intended purpose. The fountain isn't aesthetically beautiful, but you can clearly hear the sound of the water*) ⁽¹⁾.

Figure 4. SIDI DJELLIS SQUARE in THE MEDINA (OLD CITY) OF CONSTANTINE. Completely surrounded by buildings, this square is well shielded from the noise of urban traffic. It is a place that attracts people, often regulars, who perceive its particular soundscape as clearly reflecting that of an enclosed space. It is a public space with irregular features and opens to the outside through corner openings. The fountain, with its sound, shapes and modulates all the activity of the square.



(Place Sidi Djellis Constantine-Algéria Photo by the author)

While the fountain may be simple in its design, it fits well with the size and shape of the square (*there's a fountain I don't find attractive, the one at Place Sidi Djellis. But we still go there. It's very peaceful!*) ⁽²⁾. Everyone agrees on its calming and relaxing nature (*it's clear. It's a natural sound, it doesn't bother me at all*) ⁽³⁾. It is easy to feel that the presence of this fountain has brought life and movement to this confined space. (*When I come here, I feel like I'm in the countryside!*) ⁽⁴⁾. The fountain strongly defines its space.

The reflections in this enclosed area amplify its sound (The noise here is different! I could spend entire days here without a problem. It feels like a break from the overwhelming noise of the nearby boulevard, which is unbearable!) ⁽⁵⁾.

⁽¹⁾The respondents' answers are translated from Arabic

⁽²⁾The respondents' answers are translated from Arabic

⁽³⁾The respondents' answers are translated from Arabic

⁽⁴⁾The respondents' answers are translated from Arabic

⁽⁵⁾The respondents' answers are translated from Arabic



The noise in this square is less intrusive but still prominent; however, it is natural, making it more tolerable. The tranquility of the square today is not one of silence or total absence of noise, but rather the calm of natural and aesthetic sound. The fountain brings life and activity, masking the oppressive silence of the square. One could say that comfort doesn't lie in silence, but in the aesthetic quality of sound. This space, acoustically comfortable and marked by its soundscape, defines a territory where people come to find well-being and enjoyment.

3.3 RECORDING AND SOUND SEQUENCE WORK

3.3.1 Results

A phase of sound recordings was carried out simultaneously with the metrological surveys at the study sites. The sound recordings and the use of these sound sequences added a complementary dimension to the metrological work and sociological surveys conducted at each site.

A sound sequence can be as rich in auditory information as a well-taken photograph is in visual detail. Listening to and analyzing these sequences often bridges the gap between the subjective expressions from interviews and the objective data from physical measurements. Analyzing sound fragments from the same point or from different points on the site, in a particular order or another, can sometimes more effectively reveal certain sequential aspects and potential modulations of the studied sound environment, particularly highlighting the existence of sound markings and territorial boundaries in the space.

To capture such spatial modulations along a given axis of the site, the process of sound recording was divided into several static points (*measurement points*), where both recordings and acoustic measurements were taken. In this part of the study, sound marking and the perceptual concept of sound delimitation were clearly discernible in the treatment of the sound tracks.

The selection of sound fragments for recording was based on their richness and relevance in representing various aspects of sound delimitation and marking in the space. These aspects often manifested through fluctuations and masking phenomena, as well as interferences among the different noises that make up the

urban sound environment, such as the sound of passing trams, cars, people talking, and other sonic layers like the fountain noise, urban drone, and background noise. These elements gave rise to a number of clearly delineated acoustic spaces, shaped by the different manifestations of sound.

3.3.2 Discussion: case study: Place Aïn Fouara in Setif (Algeria)

The listening and analysis of sound recordings were crucial in identifying the clear presence of sound marking in the use of this central public space.

Figure 5. Fountain in Aïn Fouara Square in Sétif:
The fountain, the monument, and the water flowing from it, in both their sonic and visual aspects, create and redefine the urban space around them.



(Place Aïn Fouara, Setif-Algeria Photo by the authors)

The analysis aimed to uncover sound imprints that acoustically reflect the reality of life in the square. Sound emergences often characterize this landscape due to the various sound masking and interference phenomena between different sources present in this urban environment. Below are brief descriptions of sound sequences recorded at different points in the public square of AÏN FOUARA: The points are taken along an axis facing the fountain, from *point (1)*, the closest to the fountain, to *point (6)*, the farthest, with a distance of 3 meters between each point.

- *Point (1)*: The sound of the fountain is very perceptible. Voices from the café across the street are barely audible. Faint "vibrations" from a bus passing at a distance.



- *Point (2):* The sound of the water is still perceptible but quieter. The atmosphere of a café terrace. Conversations are easy between young people sitting on the fountain's edge. The call to prayer from a nearby mosque can be heard at 12:30 p.m. A bus passes. Cars pass and honk.
- *Point (3):* The sound of the fountain is faint but perceptible. A car drives away and a bus passes. Café conversation and ambiance.
- *Point (4):* The sound of the fountain is still perceptible. Voices are speaking, and children are shouting. A bus passes noisily, causing the fountain's sound to fade away.
- *Point (5):* The sound of the fountain is barely perceptible. Urban drone noise. Cars and buses pass by, becoming increasingly loud and frequent.
- *Point (6):* The sound of the fountain is no longer perceptible, overtaken by urban noise.

4 CONCLUSIONS

As a result of this study on the aspect of sound marking boundaries and delimitation in urban space, it can generally be said that the sound of the fountain effectively highlights the spatial dimension of a public square. This constant sound marker often provides the sensation of being in an enclosed or open space. The perception of the fountain's water sound alone is sometimes enough to convey the spatial quality of the place. The presence or absence of sound reflections clearly reveals a certain aspect of the spatiality of the location. The sound of an enclosed or open space reinforces the visual perception of the spatiality of the place by either overlaying or not overlaying the site's morphological boundaries.

- Reverberation in such spaces, for example, brings distances closer and narrows the space. This is a form of delimitation that differs from the morphological boundaries of the space. These aspects are clearly highlighted both by interviews and by the measurement of various acoustic parameters in the public space. The opposite effect (or absence of reverberation) is observed or felt in open urban spaces.
- The sound of the fountain significantly modulates the perception of space. The presence of water noise in the center of a public square creates a



disconnect between the sound or perceptual space and the physical or morphological space of the place. The nature and extent of this disconnect profoundly affect the quality and experience of the space. This is a form of spatial sound delimitation that complements the morphological one, and through overlap or layering, it creates a sense of marking boundaries in the experience and use of urban space.

- In terms of designing public spaces for comfort, the sound of a fountain could be used to modulate the perception of urban space spatiality and, in turn, adjust the boundaries shaped by sound, visual, or morphological boundaries. The experience of the urban place depends on this relationship. One practical outcome of this study is the idea that the sound of a fountain can remedy or mitigate the perception of overly enclosed or overly open spaces or correct other shortcomings in the perception and use of urban space.
- In many cases in this study, space appears clearly delimited by sound. Around a fountain in a public square, sound boundaries shaped by the relationship between water noise and urban noise, based on the classic figure-ground schema, were observed. These relationships are, in turn, modulated by different acoustic or perceptual phenomena, such as sound masking, interference, and reflection. For most people, it is more comfortable when the balance is in favor of the water noise. This is clearly reflected in the timing, frequency, and spatial extent of daily activities in urban spaces or sub-spaces where this balance is achieved. Some spaces are more frequented than others depending on the soundscape of the place. Thus, a certain delimitation is created, which is clearly evident in how people perceive, use, and live in the space. In some of the studied squares, benches and café tables were arranged in geometric patterns that clearly reflect zoning based on sound data, often forming concentric arrangements around the fountain's noise.
- The study also noted that many fountains, despite being weaker in sound intensity compared to the surrounding urban noise, manage to cover their morphological space to some extent. This phenomenon of sound marking is evident in almost all the sites studied. This has led to a reconsideration of



the concept of acoustic space in the management of urban areas. An acoustic space, whether or not it aligns with the morphological space, is an inevitable reality created by any dominant sound source in the urban soundscape. This is particularly the case in this study, where the presence of a fountain in a public square was examined. While some fountains confine their acoustic space within the morphological boundaries of the square, others, with higher sound intensity or richer spectral compositions in high frequencies, extend beyond these limits. Finally, the study revealed that some fountains, during busy times, struggle to be heard just a few meters away.

- The sound delimitation of space depends on several factors, many of which were highlighted in this study. These include aspects related to the morphology and acoustic qualities (soundscape) of the space, the nature of the fountain's noise (intensity, frequency composition, etc.), and the types of organization and composition of the surrounding urban sound environment.

5 FURTHER RESEARCH

- Further studies should be conducted on other aspects to identify research paths on elements that compose the architectural and urban atmospheres, such as lighting, sunlight, etc., which could lead to operational and functional tools for spatial and territorial marking, and possibly for the design and management of urban or public spaces.
- Further studies which aimed at developing operational tools to integrate the dimension of noise in the design of public spaces, with the objective of establishing a sound boundary for these urban areas.
- Consider studies that would aim to explore the physical and perceptual aspects of the reconfiguration of urban noise by using the sound of fountains as an acoustic mask in urban settings.
- Study of fountains as urban furniture and sound instrumentation of public spaces.



- Study the practical possibilities of reconfiguring disruptive urban sound environments by introducing various sound sources that have a positive connotation in human auditory perception.



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